3.5 Optional application 3 – discrete mathematics

3.5.1 DA: Graphs

	Content
DA1	Understand and use the language of graphs including: vertex, edge, trail, cycle, connected, degree, subgraph, subdivision, multiple edge and loop.

DA2	Identify or prove properties of a graph including that a graph is Eulerian, semi-Eulerian or Hamiltonian.

	Content
DA3	Understand and use Euler’s formula for connected planar graphs.

	Content
DA4	Use Kuratowski’s Theorem to determine the planarity of graphs.

	Content
DA5	Understand and use complete graphs and bipartite graphs, including adjacency matrices and the complement of a graph.

	Content
DA6	Understand and use simple graphs, simple-connected graphs and trees.

	Content
DA7	Recognise and find isomorphism between graphs.